

YIHUA LIU

✉ fwcnigo@gmail.com · ☎ (+86) 150-5345-8529 · 🌐 Wendy42

🎓 EDUCATION

Peking University, Beijing, China 2018.9 – 2023.7

Bachelor of Science in Data Science and Big Data Technology, Yuanpei College

Academic Honors:

- Overall GPA: 3.52/4.0; **GPA in Last Two Years: 3.77/4.0**

🏛️ EXPERIENCE

University of Houston 2025.8 – 2026.3

Research Intern

Research Area: Training-free high-resolution optimization for Diffusion Transformers (DiTs)

- **Inference Acceleration** with a **semantic region-adaptive** diffusion framework
- **Resolution Extrapolation** with a **attention distribution control method** and a frequency-based **positional embedding interpolation method**

Research Outcomes:

- Co-authored a manuscript on semantic region-adaptive inference acceleration for DiTs
- Led a first-author manuscript on resolution extrapolation for high-resolution DiTs

Wizard Quant 2023.7 – 2025.7

AI Inference Engineer

Core developer of the company's internal AI Inference Framework based on C++17, supporting **High-Frequency Trading Model** running on x86-64 CPU, which provides:

- Static Computational Graphs optimized by a series of Optimization Passes
- Highly optimized CPU kernels based on SIMD instructions
- Multi-Level benchmarking, profiling and unit testing infrastructure for production model deployment

Wangxuan Institute of Computer Technology, Peking University 2022.10 – 2023.5

Research Intern

3D Point Cloud Completion via Masked Generative Transformers

- **3D Transformer Adaptation:** Designed and implemented a novel architecture inspired by *MaskGIT* to solve 3D Point Cloud Completion task
- **Results:** Achieved competitive performance on Multi-View Partial Point Cloud Dataset, demonstrating 10% reduction in the Chamfer Distance metric (Compared with SnowflakeNet baseline)

Institute of Computational Linguistics, Peking University 2021.8 – 2022.3

Research Intern

Using BERTs for Chinese Grammatical Error Correction:

- **Model Reproduction & Verification:** Reproduced the baseline performance of the State-of-the-Art (SOTA) model initialized by the BERT Encoder on multiple datasets (**SIGHAN15**, **HybirdSet** and **TtTSet**)

- **Experiment Management:** Conducted the large-scale experimental pipeline, including data pre-processing, GPU training setup, and hyperparameter tuning to ensure stable and reproducible research outcomes

ONGOING PROJECT

LLM Agent Workflow Characterization

2026.5–Present

- Conducting an early-stage study of multi-agent workflows
- Building container-based instrumentation and trace analysis pipelines to identify opportunities for future agent runtime optimization

HONORS AND AWARDS

Work Recognition:

- Outstanding Newcomer Award, Wizard Quant, 2023
- Outstanding Project Award of the IT Department, Wizard Quant, 2024

SELECTED MANUSCRIPTS

Bowen Lin, Fanjiang Ye, Yihua Liu, Zhenghui Guo, Boyuan Zhang, Weijian Zheng, Yufan Xu, Tiancheng Xing, Yuke Wang, Chengming Zhang “*SDIT: Semantic Region-Adaptive for Diffusion Transformers*”

Yihua Liu, Fanjiang Ye, Bowen Lin, Rongyu Fang, Chengming Zhang “*TIDE: Text-Informed Dynamic Extrapolation with Step-Aware Temperature Control for Diffusion Transformers*”

SKILLS

- **Programming:** Experienced in C++ / C / Python, comfortable with Rust / Golang
- **Experiment Utils:** PyTorch, CMake, Linux, Git, LaTeX, Markdown
- **Languages:** Mandarin (Native), English (TOEFL iBT: 101)

URLS

- **Homepage:** <https://wendyl42.github.io>
- **Github:** <https://github.com/Wendyl42>
- **Linkedin:** <https://www.linkedin.com/in/yihua-l-42wdl>